

Genome-wide analysis of primary auxin-responsive *Aux/IAA* gene family in maize (*Zea mays*. L.)

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Abstract The phytohormone auxin is important in various aspects of organism growth and development. *Aux/IAA* genes encoding short-lived nuclear proteins are responsive primarily to auxin induction. Despite their physiological importance, systematic analysis of *Aux/IAA* genes in maize have not yet been reported. In this paper, we presented the isolation and characterization of maize *Aux/IAA* genes in whole-genome scale. A total of 31 maize *Aux/IAA* genes (*ZmIAA1* to *ZmIAA31*) were identified. *ZmIAA* genes are distributed in all the maize chromosomes except chromosome 2. *Aux/IAA* genes expand in the maize genome partly due to tandem and segmental duplication events. Multiple alignment and motif display results revealed major maize *Aux/IAA* proteins share all the four conserved domains. Phylogenetic analysis indicated *Aux/IAA* family can be divided into seven subfamilies. Putative *cis*-acting regulatory DNA elements involved in auxin response, light signaling transduction and abiotic stress adaption were observed in the promoters of *ZmIAA* genes. Expression data mining suggested maize *Aux/IAA* genes have temporal and spatial expression pattern. Collectively, these results will provide molecular insights into the auxin metabolism, transport and signaling research.

Keywords Maize · Auxin · *Aux/IAA* gene family · Bioinformatic analysis

Introduction

Plant hormone auxin/indole-3-acetic acid (*Aux/IAA*) is vital in diverse cellular and developmental processes, including embryogenesis, lateral root initiation, axis formation and patterning, vascular elongation, leaf expansion, inflorescence and fruit development, tropism, apical dominance, defense responses against pathogens, etc. [1–3]. At an early stage of auxin signaling transduction, some gene families, such as *Aux/IAA*, *GH3* (*Gretchen Hagen3*) and *SAUR* (*small auxin up RNA*) are responsive to auxin stimulation. Thus, these gene family members are termed primary/early auxin-responsive genes [4].

Aux/IAA gene was firstly isolated in soybean [5]. Canonical *Aux/IAA* proteins share four characterized regions designated domain I, II, III and IV. Domain I in N terminal is with the function of transcriptional repression. Domain II is related with protein stability. Domain III and IV is responsible for protein dimerization (homo- and heterodimerization). Homodimerization refers interactions between *Aux/IAA* gene family members. Heterodimerization is the interplay of *Aux/IAA* and ARF (auxin response factor) proteins [6, 7].

Generally, *Aux/IAA* as nucleus-localized and short-lived proteins are degraded easily. There are still some long-lived *Aux/IAA* proteins which are not degraded easily [8]. Auxin-induced degradation of *Aux/IAA* protein is required for the SCF^{TIR1} ubiquitin ligase complex. Moreover, *Aux/IAA* protein turnover mediated by ubiquitin is proteasome dependent and indispensable to auxin response.

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The degradation process can be delayed effectively by modifying the domain II of Aux/IAA proteins [9, 10].

Aux/IAA genes are induced rapidly by auxin employment. Besides, *Aux/IAA* genes, as important regulators of auxin-modulated genes expression, can trigger (activation or repression) the auxin-mediated transcription [11]. The auxin response suppression behavior is beneficial to form gradients of auxin concentration which are necessary for many organs and tissues formation [12]. Intriguingly, the activity of Aux/IAA protein is modulated by phytochrome A phosphorylation. In this condition, light and auxin signaling pathways are integrated through Aux/IAA proteins [13]. Moreover, interaction of brassinosteroid and auxin signaling pathways has been observed [14].

Sequences derived from large-scale sequencing projects are informative in functional genomics research and provide the opportunity to genome-wide scan of gene families. The B73 maize genome sequencing project was initiated in 2005 [15]. Results of B73 genome sequencing were published in Science magazine on November 20, 2009 [16]. Filtered gene sets (version 4a.53) opened publicly in June 2009. Based on sequence resources of the maize genome, we aimed at investigating maize *Aux/IAA* genes in whole-genome scale. Maize *Aux/IAA* genes number, their genomic organization, expansion pattern, motif distribution, phylogenesis, *cis*-elements in promoter regions and expression profiles were analyzed in detail. Results presented in this study will provide molecular insights into the auxin signaling pathway research.

Materials and methods

Isolation of *Aux/IAA* genes in maize

The accession number of *Aux/IAA* gene family in the Pfam is PF02309 (<http://pfam.sanger.ac.uk/>) [17]. The HMM (hidden Markov model) profile of *Aux/IAA* family was extracted from the Pfam. Based on the HMM profile, 261 aa (amino acid) conserved sequences of Aux/IAA proteins were obtained in aid of HMMEMIT utility [18]. The above 261 aa conserved sequences were adopted to query the B73 maize sequencing database (<http://www.maizesequence.org/index.html>). Searching parameters were as followings: BLASTp, database—filtered gene sets, *E*-value < e^{-10} , other parameters were defaulted. Besides, the NCBI database (<http://www.ncbi.nlm.nih.gov/>) was mined using IAA as a keyword to avoid missing other maize *Aux/IAA* genes. Redundant sequences were removed. Then, the Pfam, SMART (<http://smart.embl-heidelberg.de/>) and InterProScan (<http://www.ebi.ac.uk/Tools/InterProScan/>) web server were used to examine the conserved domains of Aux/IAA proteins [17, 19, 20]. After being examined, non-

redundant sequences were employed to search the maize sequencing and NCBI databases constantly till no other Aux/IAA proteins were retrieved. Basic physical and chemical parameters of maize *Aux/IAA* genes were calculated by online ProtParam tool (<http://www.expasy.org/tools/protparam.html>).

Gene structure analysis of maize *Aux/IAA* genes

The information of maize *Aux/IAA* genes, including accession number, chromosomal location, ORF length, exons and introns number, were retrieved from the B73 maize sequencing database. FLcDNA (full length cDNA) sequences of maize *Aux/IAA* genes were obtained from the maize FLcDNA project website (<http://www.maizecna.org/>). Structures of maize *Aux/IAA* genes were showed by GSDS tool (<http://gsds.cbi.pku.edu.cn/>) [21].

Motif display and phylogenetic analysis of Aux/IAA proteins

MEME utility was used to display motifs of Aux/IAA proteins from maize, rice and *Arabidopsis* (http://meme.nbcr.net/meme4_1/cgi-bin/meme.cgi) [22]. Parameters were set as followings: the occurrences of a single motif—zero or one per sequence, motifs width ranges from 10 to 300 aa, maximum number of motifs to find—5, other parameters were defaulted. Multiple alignment of Aux/IAA proteins were conducted by Clustal X software with defaulted parameters. Phylogenetic analysis was performed with Neighbor-Joining method and 1,000 bootstrap trials using Clustal X tool in conjunction with MEGA software [23, 24].

Expansion pattern research of maize *Aux/IAA* family

Tandem duplication and segmental replication have effect on gene family amplification. In this study, two genes from one species which located in the same clade of the phylogenetic tree were called co-paralogs. Co-paralogs are regarded as tandem duplicates on condition that five or fewer protein-coding genes between them [25]. For segmental replication event investigation, twenty protein-coding genes flanking co-paralogs each (ten genes upstream and downstream of every member of co-paralogs) were retrieved based on the B73 maize genome annotation results. Co-paralogs are considered to be segmental duplicates provided that there are co-paralogs in neighboring protein-coding genes [26].

Promoter regions analysis of *ZmIAA* genes

To investigate *cis*-elements in promoter sequences of maize *Aux/IAA* genes, 2,000 bp of genomic DNA sequences

upstream of initiation codon (ATG) were downloaded from the B73 maize sequencing database. The PLACE website (<http://www.dna.affrc.go.jp/PLACE/>) was adopted to identify *cis*-elements in the promoters [27].

Expression profiles investigation of maize *Aux/IAA* genes

Transcript levels of maize *Aux/IAA* genes were analyzed by multiple methods. Firstly, EST mining was performed in the NCBI EST database (<http://www.ncbi.nlm.nih.gov/dbEST/>) using megablast tool. Parameters of searching were as followings: maximum identity > 95%, length > 200 bp and E value < 10^{-10} . Secondly, expression data in the PlantGDB and MAGI databases, including EST, cDNA and PUTs (PlantGDB unique transcripts), were retrieved by the MaizeGDB genome browser tool (<http://gbrowse.maizegdb.org/gbrowse/cgi-bin/gbrowse/maize>).

Results

Isolation and identification of *Aux/IAA* genes in maize

The 261 aa conserved sequences of *Aux/IAA* proteins generated from HMM profile in the Pfam were employed to search the B73 maize sequencing database. Additionally, the NCBI database was queried in keyword “IAA” lest some maize *Aux/IAA* genes were missed. A total of 161 sequences were obtained, including 82 non-redundant sequences. Then, fifty-two sequences were confirmed to contain conserved domains of *Aux/IAA* proteins. Of these fifty-two sequences, some were different deduced transcripts of one genomic sequence. In this study, only the transcript with most conserved domains, that is, the transcript is with lowest E-value of domain examination, was selected. Sequences which were not listed in the B73 maize filtered gene sets were also eliminated. The remaining sequences were adopted to search the maize sequencing and NCBI databases continuously till no other maize *Aux/IAA* genes were found. Finally, thirty-one maize *Aux/IAA* genes were obtained. These genes were designated sequentially from *ZmIAA1* to *ZmIAA31* according to their genome location (Table 1).

Characters of *ZmIAA* genes

Thirty-one *ZmIAA* genes were among all the maize chromosomes except chromosome 2. The number of genes per chromosome ranged from one to seven (Fig. 1). Schematic structures of *ZmIAA* genes were showed by GSDS utility in Fig. 2. Average intron number of *ZmIAA* genes was three.

According to number of introns, thirty-one *ZmIAA* genes could be divided into three groups. Two genes *ZmIAA22* and *ZmIAA31* belonged to intronless genes. Eleven genes had one or two introns. Intron number of the rest was with a range of three to five. Additionally, some physical and chemical parameters of maize *Aux/IAA* proteins, including molecular weight, isoelectric point, etc., were estimated by ProtParam tool. Negative GRAVY (grand average of hydropathicity) index indicated that maize *Aux/IAA* proteins were all hydrophilic. Only five maize *Aux/IAA* proteins *ZmIAA13*, *ZmIAA20*, *ZmIAA24*, *ZmIAA25* and *ZmIAA26* with stability index less than 40 might be stable in vitro (Supplementary Table 1).

Multiple alignment results showed that major maize *Aux/IAA* proteins shared four characterized regions designated domain I, II, III and IV. Two NLSs (nuclear localization signal) which are related with the nucleus location of *Aux/IAA* proteins were identified. The $\beta\alpha\alpha$ motif (one β sheet and two α helices) with an important role in dimerization of *Aux/IAA* proteins was also found. Moreover, phosphorylation sites of multiple kinases were observed in maize *Aux/IAA* proteins. The existence of multiple phosphorylation sites partly implied that maize *Aux/IAA* proteins belong to short-lived proteins (Supplementary Fig. 1).

Motif distribution and phylogenesis of *Aux/IAA* proteins

MEME web server was used to analyze motif distribution in maize, rice and *Arabidopsis* *Aux/IAA* proteins. Four conserved domains of *Aux/IAA* proteins showed in Supplementary Fig. 1 were divided into five motifs by MEME tool. Motif 2 and 4 constituted the longest domain IV. Domain I, II and III were represented by motif 5, 3 and 1, respectively (Table 2). Motif 1 existed in all the *Aux/IAA* proteins. Motif 2 was absent in four maize *Aux/IAA* proteins. Motif 3 was not observed in eighteen *Aux/IAA* proteins. Interestingly, *AtIAA32* protein contained only one motif (Fig. 3).

To investigate the relationships of *Aux/IAA* proteins, thirty-one maize, thirty-one rice and twenty-nine *Arabidopsis* *Aux/IAA* proteins were adopted to build the phylogenetic tree. Ninety-one *Aux/IAA* proteins could be classified into seven groups. Twelve pairs of maize/rice *Aux/IAA* proteins were in the same clade of the phylogenetic tree. By contrast, no maize/*Arabidopsis* *Aux/IAA* proteins positioned in the same subtree of the dendrogram (Supplementary Fig. 2).

Genome duplication of *ZmIAA* genes

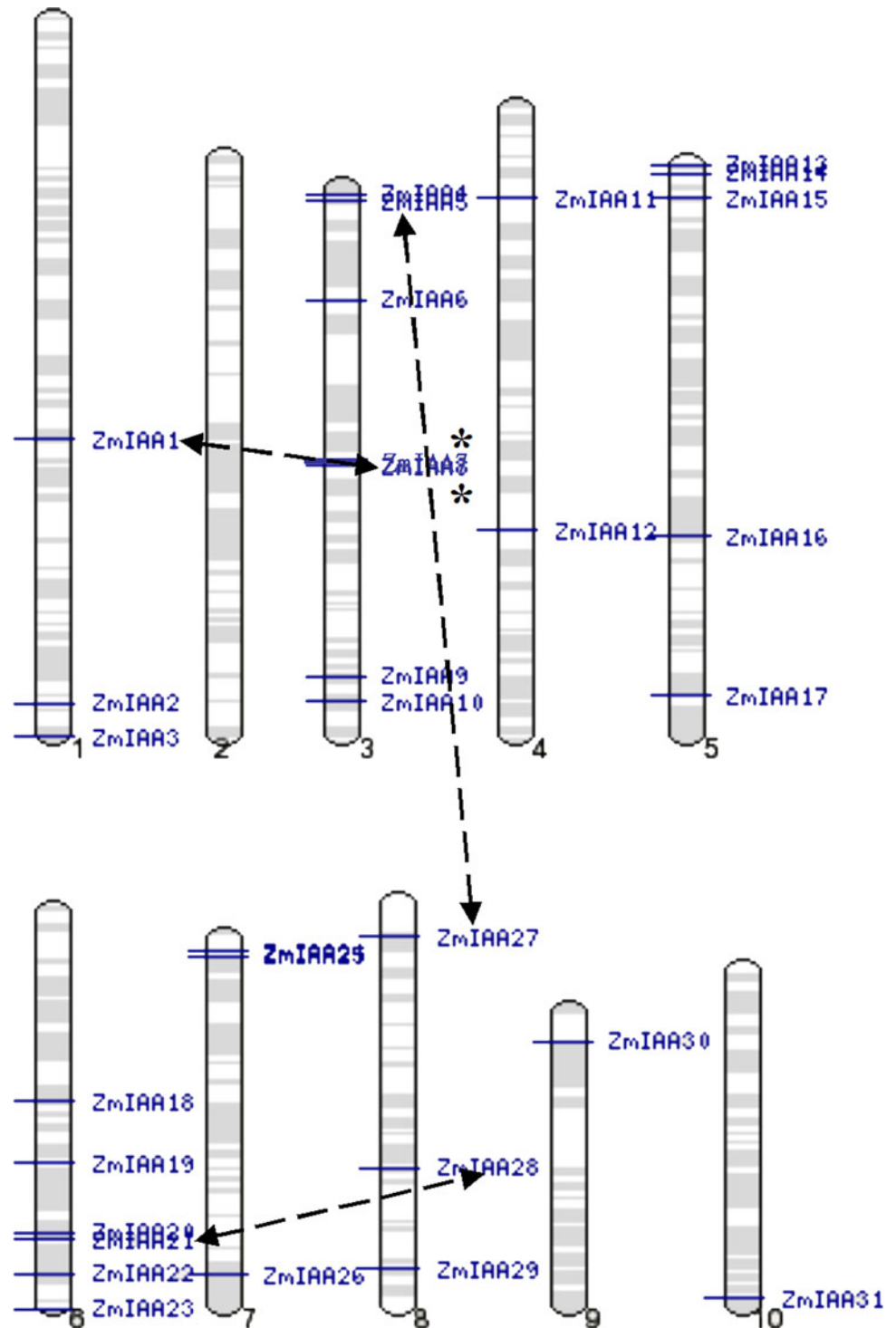
Based on phylogenetic results presented in Supplementary Fig. 2, nine co-paralogs were identified in *ZmIAA* genes.

Table 1 Characters of *Aux/IAA* genes of maize

Gene	Accession number		Genome location		Chr (Bin)	ORF (bp)	Exon number	Intron number
	Ensembl transcript	Maize F1cDNA	Clone number	Location				
<i>ZmIAA1</i>	GRMZM2G079957_P02	BT061793	AC209175.3	170401374–170403227	1.06	681	3	2
<i>ZmIAA2</i>	GRMZM2G159285_P01	BT055925	AC186412.4	274031458–274034225	1.11	1185	5	4
<i>ZmIAA3</i>	GRMZM2G013158_P01	BT062744	AC195205.2	287308744–287310492	1.12	777	3	2
<i>ZmIAA4</i>	GRMZM2G104176_P01	BT069155	AC190549.3	6923951–6926589	3.02	687	6	5
<i>ZmIAA5</i>	GRMZM2G004696_T01	BT040328	AC210083.2	9864760–9868114	3.03	660	4	3
<i>ZmIAA6</i>	GRMZM2G074742_P01	NA	AC191649.3	48580945–48582725	3.04	594	4	3
<i>ZmIAA7</i>	GRMZM2G138268_P01	BT043328	AC190621.2	113988619–113993224	3.05	813	5	4
<i>ZmIAA8</i>	GRMZM2G167794_P01	BT083711	AC212335.3	114260768–114262484	3.05	690	3	2
<i>ZmIAA9</i>	GRMZM2G057067_P01	BT069349	AC212001.2	197512524–197515012	3.08	1071	6	5
<i>ZmIAA10</i>	GRMZM2G037368_P01	BT041657	AC191590.3	207539202–207542564	3.09	807	5	4
<i>ZmIAA11</i>	GRMZM2G059544_P01	BT065805	AC194950.3	39400756–39401986	4.04	699	3	2
<i>ZmIAA12</i>	GRMZM2G142768_P01	BT087785	AC186507.4	170463924–170466346	4.07	879	5	4
<i>ZmIAA13</i>	GRMZM2G152796_P01	BT063876	AC191048.2	4190251–4191678	5.01	543	3	2
<i>ZmIAA14</i>	GRMZM2G077356_P01	BT069632	AC185441.3	7732528–7735474	5.01	684	5	4
<i>ZmIAA15</i>	GRMZM2G128421_P01	BT055320	AC198181.4	17228447–17229880	5.03	672	3	2
<i>ZmIAA16</i>	GRMZM2G121309_P01	BT061169	AC190547.3	151037634–151041541	5.04	867	6	5
<i>ZmIAA17</i>	GRMZM2G030465_P01	NA	AC211323.4	213610225–213611459	5.08	618	3	2
<i>ZmIAA18</i>	GRMZM2G000158_P04	BT068319	AC205727.3	78755749–78757496	6.02	591	4	3
<i>ZmIAA19</i>	GRMZM2G079200_P01	BT083502	AC194137.3	103265305–103267002	6.04	594	4	3
<i>ZmIAA20</i>	GRMZM2G051769_P01	BT041580	AC194007.2	130267317–130271766	6.05	528	3	2
<i>ZmIAA21</i>	GRMZM2G147243_P02	BT033707	AC205900.3	133397528–133401260	6.05	732	2	1
<i>ZmIAA22</i>	GRMZM2G141205_P01	NA	AC206058.3	146671458–146672153	6.05	693	1	0
<i>ZmIAA23</i>	GRMZM2G074427_P02	NA	AC206059.3	160288650–160292150	6.08	1038	6	5
<i>ZmIAA24</i>	GRMZM2G149449_P01	BT040293	AC214119.3	9279154–9279875	7.01	345	3	2
<i>ZmIAA25</i>	GRMZM2G115357_P01	BT084529	AC204533.3	10854843–10856169	7.01	606	3	2
<i>ZmIAA26</i>	GRMZM2G048131_P01	BT094005	AC193413.3	137198544–137200001	7.04	417	4	3
<i>ZmIAA27</i>	GRMZM2G130953_P02	BT034414	AC211862.4	18360850–18362941	8.02	558	4	3
<i>ZmIAA28</i>	GRMZM2G035465_P03	BT069580	AC201782.4	109479256–109482884	8.04	768	5	4
<i>ZmIAA29</i>	GRMZM2G163848_P05	BT061665	AC187246.5	149197736–149200798	8.06	816	5	4
<i>ZmIAA30</i>	GRMZM2G001799_P01	BT069772	AC200401.4	16151066–16152954	9.01	648	4	3
<i>ZmIAA31</i>	GRMZM2G134517_P01	BT064184	AC203327.3	133876405–133877172	10.07	765	1	0

NA not available

Fig. 1 Chromosome distribution and expansion pattern of maize *Aux/IAA* genes. Thirty-one *Aux/IAA* genes are among maize 10 chromosomes except chromosome 2. Tandem duplicates *ZmIAA7/ZmIAA8* in chromosome 3 were emphasized by asterisks. Segmental duplicates, including *ZmIAA1/ZmIAA8*, *ZmIAA5/ZmIAA27*, *ZmIAA21/ZmIAA28*, were indicated by bidirectional arrows in figure



Twenty protein-coding genes flanking co-paralogs each (ten proteins in each side of every member of co-paralogs) were retrieved from the B73 maize sequencing database. According to the B73 maize genome annotation results, co-paralogs *ZmIAA7/ZmIAA8* separated by two protein-coding genes were considered to be tandem duplicates.

“New” co-paralogs were found in protein-coding genes neighboring three co-paralogs *ZmIAA1/ZmIAA8*, *ZmIAA5/ZmIAA27* and *ZmIAA21/ZmIAA28*. Thus, these three co-paralogs could be segmental duplicates (Fig. 1). Other five co-paralogs may generate owing to different duplication events, such as random duplication and/or insertion events.

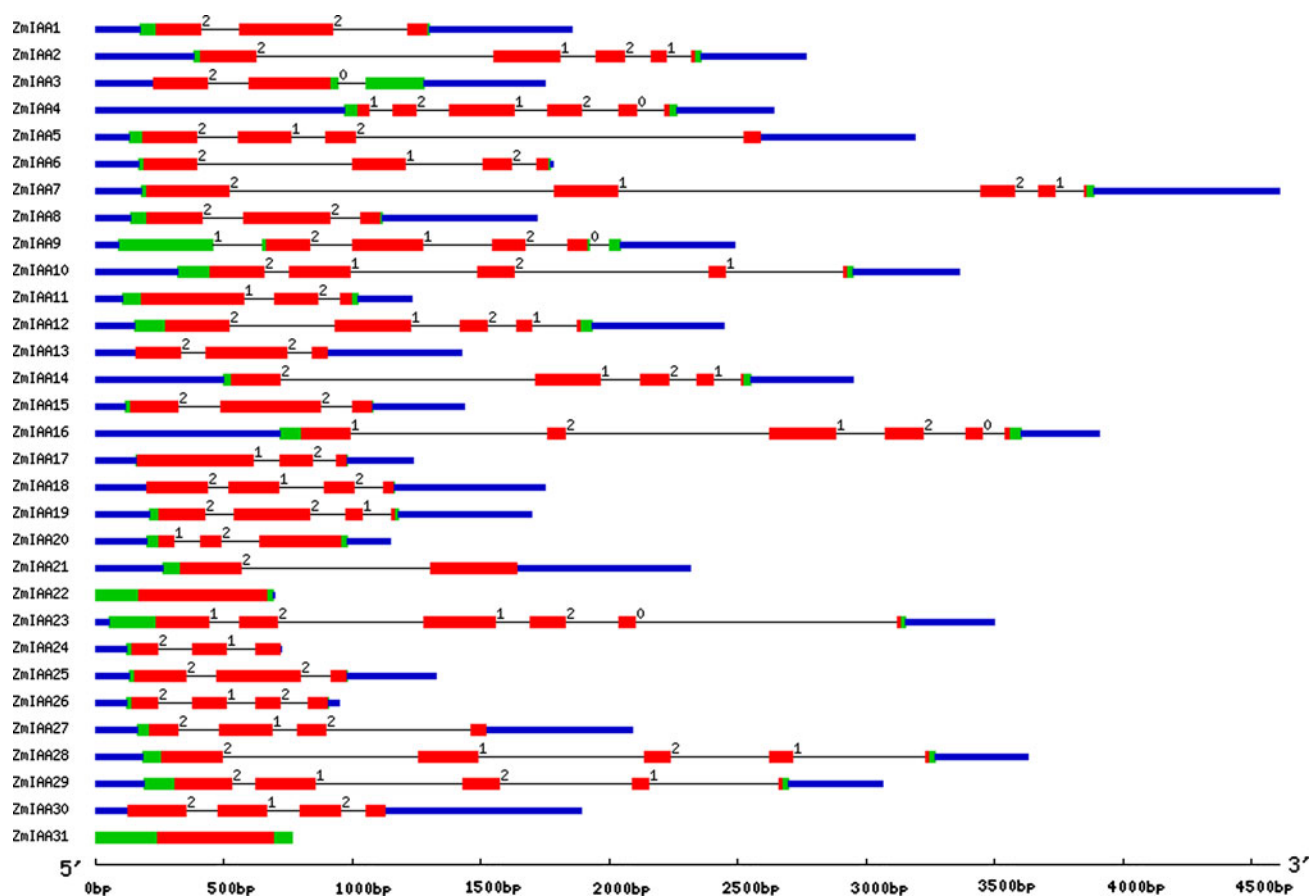


Fig. 2 Gene structure of maize *Aux/IAA* family. Exons and introns were showed by filled boxes and single lines, respectively. Conserved sequences of *Aux/IAA* proteins were marked in red. UTRs were

displayed by thick blue lines at both ends. Introns phases 0, 1 and 2 were indicated by numbers 0, 1 and 2 in figure

Table 2 Motif distribution in *Aux/IAA* proteins

Motif name	Length (aa)	Sequence	Domain
Motif1	33	FVKVSM DGAPYL RKVD LKAYKSYDELSFALEKM	III
Motif2	22	SEYVLTYEDKDGDWMLVG DVPW	IV
Motif3	29	AAPPAAKAQ VVGW PPVRSYRK NLLASS SS	II
Motif4	21	EMFVSSCKRLRIMKGSE IGL	IV
Motif5	20	LELKATE LRLGL PGSSSSAS	I

Note: The position of *Aux/IAA* domain was mentioned in bold emphasis

cis-elements in promoter sequences of *ZmIAA* genes

For promoter regions investigation, 2,000 bp genomic DNA sequences upstream of start codon of *ZmIAA* genes were analyzed. Four types of *cis*-elements which are directly involved in auxin signaling transduction were identified in PLACE workspace. Two types of *cis*-elements with signal ID S000270 and S000026 belong to AREs (auxin response element), which are specifically binded with ARF in auxin response [28]. Other two types of *cis*-elements which are required for auxin response were also found in the promoters of *ZmIAA* genes (Fig. 4).

Expression pattern of maize *Aux/IAA* family

EST data can provide valuable information for gene expression research. There are 2,018,634 maize ESTs in the NCBI EST database. The number of maize ESTs takes third place in the NCBI EST database, inferior to human and mouse ESTs number (dbEST release 090409). Thus,

Fig. 3 Motif distribution in maize, rice and *Arabidopsis* *Aux/IAA* proteins. Motifs of *Aux/IAA* proteins were investigated by MEME web server. Five motifs representing four domains I, II, III and IV were not observed in all *Aux/IAA* proteins

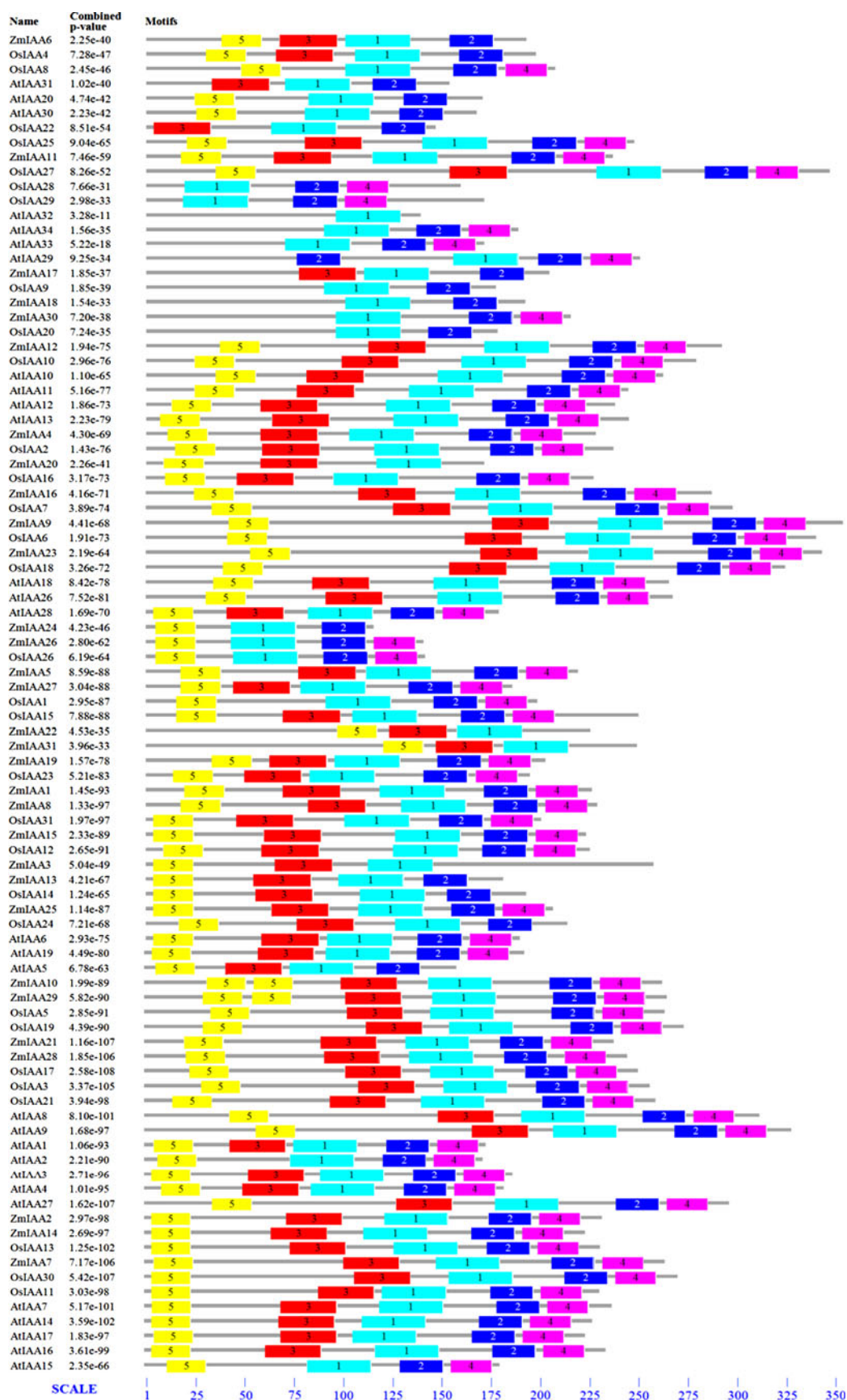
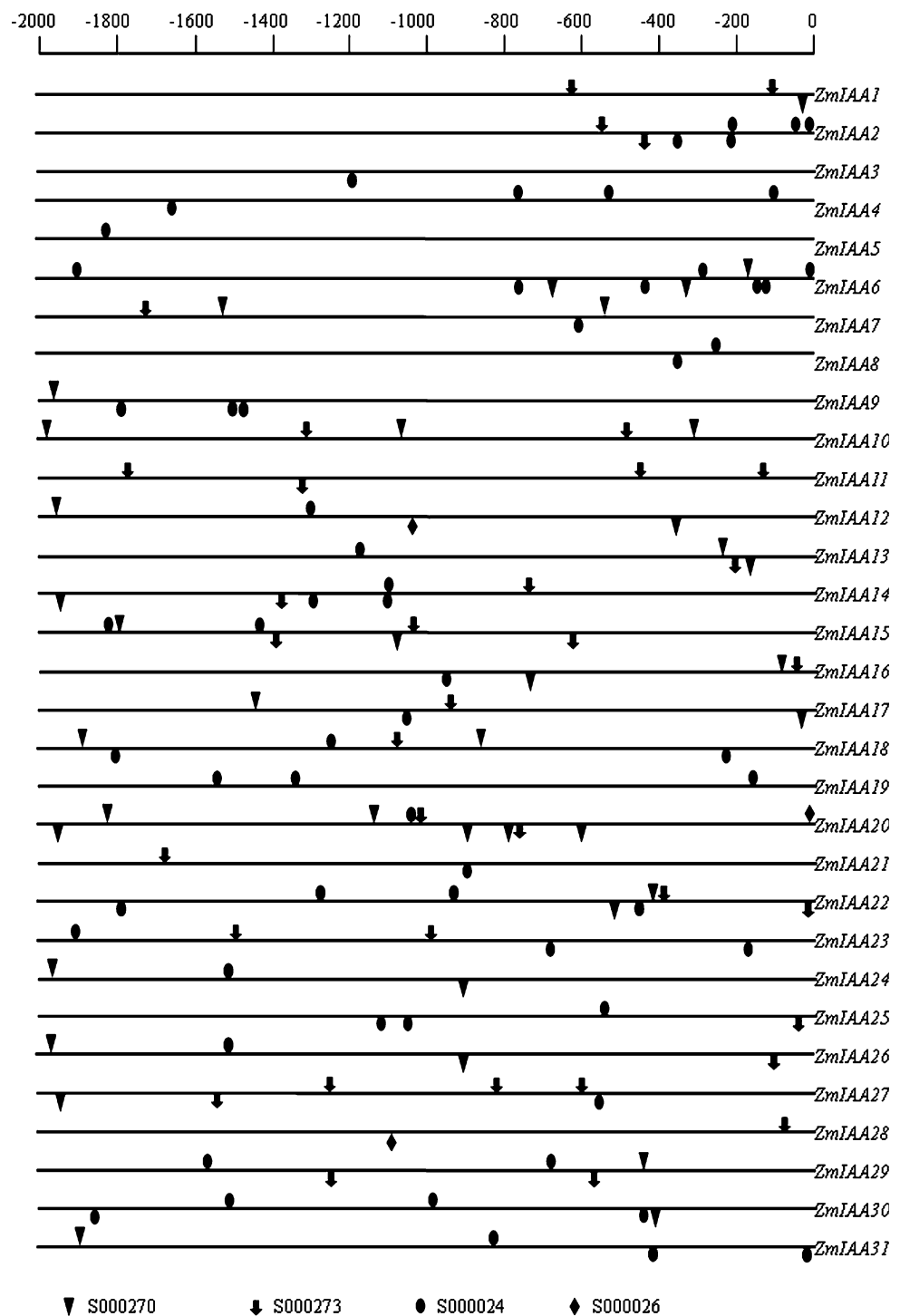


Fig. 4 *Cis*-elements in the promoters of *ZmIAA* genes. *Cis*-elements with signal ID S000270 and S000026 belong to AREs, which were indicated by triangles and diamonds, respectively. *Cis*-elements with signal ID S000273 and S000024 are necessary for auxin response, which were showed by drop-down arrows and ellipses in figure, respectively



EST mining was firstly adopted to analyze transcript levels of *ZmIAA* genes. The coding sequences of *ZmIAA* genes were used to query the est_others database using megablast tool. Based on tissue and organ types, obtained EST data were classified into nine groups (Table 3). A large number of ESTs in the NCBI EST database were generated from the maize FLcDNA project, EST data of mixed tissues and

organs released by the maize FLcDNA project were not eliminated in this study (column 10 in Table 3). On this condition, expression evidences were detected in all the *ZmIAA* genes. If mixed EST data were not taken into account, no expression evidence of *ZmIAA30* was detected. Expression data of seven *ZmIAA* genes, including *ZmIAA3*, *ZmIAA6*, *ZmIAA13*, *ZmIAA17*, *ZmIAA19*, *ZmIAA20* and

Table 3 Expression analysis of *ZmIAA* genes in silico

Gene	Tissue and organ type (NCBI)									Number of ESTs in dbEST	MAGI	PlantGDB		
	Suspension culture	Callus	Meristem	Seedling	Root	Tassel	Ear	Seed	Multiple			EST	cDNA	PUTs
<i>ZmIAA1</i>			+		+				+	24				
<i>ZmIAA2</i>		+	+		+	+		+	+	57	+			
<i>ZmIAA3</i>			+						+	6				
<i>ZmIAA4</i>	+				+	+			+	35				
<i>ZmIAA5</i>		+	+	+	+		+	+	+	89				
<i>ZmIAA6</i>								+	+	4				
<i>ZmIAA7</i>			+	+	+			+	+	102				
<i>ZmIAA8</i>			+		+		+		+	18		+		+
<i>ZmIAA9</i>					+			+	+	14	+			
<i>ZmIAA10</i>			+					+	+	30	+	+	+	+
<i>ZmIAA11</i>						+		+	+	9		+		
<i>ZmIAA12</i>			+		+				+	67	+	+		+
<i>ZmIAA13</i>			+						+	11				
<i>ZmIAA14</i>	+				+	+		+	+	48				
<i>ZmIAA15</i>	+				+			+	+	43				
<i>ZmIAA16</i>			+		+	+			+	16		+		+
<i>ZmIAA17</i>					+				+	8		+		+
<i>ZmIAA18</i>	+		+					+	+	28				
<i>ZmIAA19</i>					+				+	23	+			
<i>ZmIAA20</i>						+			+	9		+		+
<i>ZmIAA21</i>	+		+		+	+	+	+	+	137				
<i>ZmIAA22</i>			+		+			+	+	15	+			
<i>ZmIAA23</i>			+		+				+	6				
<i>ZmIAA24</i>		+						+	+	8	+		+	
<i>ZmIAA25</i>	+		+						+	50				
<i>ZmIAA26</i>	+							+	+	4				
<i>ZmIAA27</i>			+	+	+		+		+	51		+		
<i>ZmIAA28</i>			+	+	+		+	+	+	109	+	+	+	+
<i>ZmIAA29</i>			+		+			+	+	83				
<i>ZmIAA30</i>									+	2				
<i>ZmIAA31</i>			+						+	2		+	+	+

+ Expressed, *blank* not expressed

ZmIAA31, were checked in only one tissue or organ. Additionally, expression evidences of some *ZmIAA* genes were supported by transcript data in the MAGI and PlantGDB databases (Table 3).

Discussion

Aux/IAA gene family in nature

Increasing evidence supports that gene families are essential in many metabolic and developmental processes [29, 30]. Accordingly, a number of methods have been

developed to isolate gene family members. For instance, a combination of suppression subtractive hybridization and in silico cloning has been employed to obtain maize *FtsH* gene family [31]. Degenerated and gene-specific primers designed on conserved domains have been used to isolate CRY and PPR protein families [32, 33]. In this study, we isolated *Aux/IAA* genes of maize in silico.

Aux/IAA gene families only exist in plant kingdoms in nature [12]. So far, twenty-nine, thirty-one and thirty-five *Aux/IAA* genes in the *Arabidopsis*, rice and *Populus* genome have been reported [34–36]. Thirty-one *Aux/IAA* genes of maize were presented in this study. Moreover, 261 aa conserved sequences of *Aux/IAA* proteins were queried

in the NCBI non-redundant protein database. Results showed that the number of *Aux/IAA* genes in the grape and soybean genome took first and second place, respectively. Plants grouped into Liliopsida had less *Aux/IAA* genes (data not shown). Species divergence may account for the phenomenon. Certainly, data abundance in GenBank may also result in the limited number of *Aux/IAA* genes in Liliopsida.

Diverse *cis*-elements in the promoters of *ZmIAA* genes

Four types of *cis*-elements which are directly involved in auxin signaling pathway were identified in this study (Fig. 4). In view of *Aux/IAA* proteins as linkers between light perception and auxin response [13], twenty-eight types of putative *cis*-elements responsive to light stimulation were observed in the promoters of *ZmIAA* genes (Table S2). Besides, auxin can play as regulators in plant defense response [3]. Four types of *cis*-elements relating to pathogen resistance were also found in the promoter sequences of *ZmIAA* genes. Additionally, *cis*-elements which are involved in abiotic stresses adaption, such as DRE (dehydration-responsive element), LTRE (low temperature-responsive element), were also observed in the promoter regions of *ZmIAA* genes. Diverse *cis*-elements in the promoters of *ZmIAA* genes partly demonstrate that auxin signaling transduction can interact with other metabolic pathways. Crosstalk between auxin and abiotic stresses signaling pathways has been confirmed in rice [37]. Undoubtedly, the diversity of *cis*-elements in the promoters of *ZmIAA* genes is also related with the key role of *Aux/IAA* proteins in plant life.

Expression profiles and fates of *ZmIAA* genes

Expression profiles of *ZmIAA* genes were investigated by multiple strategies in this study. EST mining results indicated that major *ZmIAA* genes expressed in checked tissues and organs. However, expression evidences of some *ZmIAA* genes were detected in only one tissue or organ. Intriguingly, if mixed expression data were ruled out, no expression data of *ZmIAA30* was found in the NCBI EST, MAGI and PlantGDB databases. Many reasons may explain the phenomenon. One hypothesis is that some *ZmIAA* genes have temporal and spatial expression pattern. Tissue-specific and developmental stage-specific expression manners of *Aux/IAA* genes have been demonstrated by real-time PCR and microarray analysis in rice [35, 37]. In view of partial motifs absence and distinct expression pattern, others believed that some *ZmIAA* genes belong to pseudogenes. In *Arabidopsis*, four *Aux/IAA* genes were considered to be pseudogenes [12]. In such a scenario,

some *Aux/IAA* genes may be lost from the maize genome owing to dilution effect in selective sweep.

During past, our knowledge of auxin action mainly resulted from *Arabidopsis* mutants research. To date, there are still many unanswered questions relating to the mechanism of auxin signaling transduction [12]. Crosstalk between auxin and other signaling pathways increases the complexity of auxin signaling research. Isolation and characterization of genes involved in auxin response in genome scale may pave the way for auxin pathway investigation. Results presented in this study will lay a foundation for elucidating the complicated auxin signaling transduction cascade.

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